Topics in CS: Problem Set 7

Due date: December 28, 2025.

Question 1. (10 points) Consider the key-agreement protocol of Diffie-Hellman over the multiplicative group $\mathbb{Z}_{23}^*$ generated by the element 5. Say that Alice's secret is $a = 8$ and Bob's secret is $b = 11$.

What would be the transcript of the protocol and what would be their shared key?

Question 2. (40 points) Consider the key-agreement protocol of Diffie-Hellman over the multiplicative group $\mathbb{Z}_p^*$ for a large prime p. Notice that this is not a prime-order group since its order $p - 1$ is even. Prime-order groups are useful for two reasons:

- They provide stronger security as the discrete-log problem in a group can be reduced to solving it in all of the prime-order subgroups (we may see this later in the course).
- It is simple to find generators for prime-order groups.

In this question, we will see how to address these issues by working with primes of a special form.

1. (a) Consider the prime $11 = 2 \cdot 5 + 1$. Create a table where in the first column you write every element $x \in \mathbb{Z}_{11}^*$ and in the second column $x^2 \bmod 11$. Can you identify a subgroup of $\mathbb{Z}_{11}^*$ that has order 5 and a generator for this subgroup?
   (b) As we saw, since $\mathbb{Z}_{11}$ is a field, there is only one element of order 2; which element is it?
   (c) What can you say about the order of the elements that are not in the subgroup you identified and do not have order 2?
   (d) Consider the function $f : \mathbb{Z}_{11}^* \to \mathbb{Z}_{11}^*$ defined by $f(x) = x^5 \bmod 11$.
   Is this a homomorphism? What is the kernel and what is the image of f?
2. Let p be a prime of the form $p = 2q + 1$ where q is also a prime.
   What are the possible orders of the subgroups of $\mathbb{Z}_p^*$?
3. Prove that the subset $\{x^2 \bmod p \mid x \in \mathbb{Z}_p^*\}$ forms a subgroup of prime order q as follows:
   (a) Define a function $f : \mathbb{Z}_p^* \to \mathbb{Z}_p^*$ in the spirit of Item 1d and prove that f is homomorphism. Hint: FLT and other observations from the lecture on primality testing can be used here.
   (b) Prove that $G = \{x^2 \bmod p \mid x \in \mathbb{Z}_p^*\}$ is the kernel of f.
   (c) What is the image of f? Use the first isomorphism theorem to compute the index of G.
   (d) Using index of G and the order of $\mathbb{Z}_p^*$, compute the order of G?

   The subgroup G is called the subgroup of *quadratic residues* in $\mathbb{Z}_p^*$.
4. Given an element $x \in \mathbb{Z}_p^*$, describe an efficient way to verify whether x is a quadratic residue.

Question 3. (20 points) Say that Alice and Bob wish to communicate over an unauthenticated channel (meaning that when they receive a message, they do not know the identity of the sender), and they plan to use the key-agreement protocol of Diffie-Hellman to establish a private communication channel. However, in class we only showed security against passive eavesdroppers that can simply listen to the communication channel.

Show that the protocol is not secure against active man-in-the-middle attacks that can intercept and inject messages to the communication channel. Show that

Question 4. (30 points) Implement the Diffie-Hellman key-agreement protocol.

To generate the group sample a prime q of 512-bits, such that $p = 2q + 1$ is also prime. We will work in the subgroup of order q of quadratic residues. Compute the following steps and display these values in your output.

1. Find a generator according to Q2.
2. To play Alice's role, sample $a \leftarrow \mathbb{Z}_q$ and compute $g^a \bmod p$.
3. To play Bob's role, sample $b \leftarrow \mathbb{Z}_q$ and compute $g^b \bmod p$.
4. Compute the shared key according to Alice.
5. Compute the shared key according to Bob.

Good luck!